Fractional knapsack

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Topics covered are:

- Fractional Knapsack

INTRODUCTION

- **The Knapsack Problem**
  - **In the knapsack problem, where we are given a set S of n items, s.t., each item *i* has a positive benefit *bi* and a positive weight *wi*, and we wish to find the maximum-benefit subset that doesn't exceed a given knapsack capacity W.**

  - In the fractional knapsack problem, We are allowed to take arbitrary fractions of each item. i.e., we can take an amount x_i of each item i such that

$$0 \le x_i \le w_i \text{ for each } i \in S \quad \text{and} \quad \sum_{i \in S} x_i \le W.$$

INTRODUCTION

- **The Objective**
  - To obtain a filling of the knapsack that maximizes the total profit earned. As the knapsack capacity is W, we require the total weight of all objects to be at most W. Formally, the problem is stated as:

$$maximize \sum_{1 \le x_i \le n} P_i x_i$$

Subject to, $\sum_{1 \le x_i \le n} w_i x_i \le W$

Where, $0 \le x_i \le 1 \quad and \quad 1 \le i \le n$

INTRODUCTION

- **<u>Example,</u>**
- Objects(i):
- Profits (P):
- Weights (w):

1	2	3	4	5	6	7
5	10	15	7	8	9	4
1	3	5	4	1	3	2

- Knapsack Capacity (W) = 15
- No of items (n) = 7

Feasible solutions:

1. Fill the knapsack by collecting maximum profit items
2. Fill the knapsack by collecting minimum weight items
3. Fill the knapsack according to the ration of weight and profit of items

Feasible Solution: 1

- **Choose the items according to their maxmimum profit**

Objects	Profits	Item weights	Remaining weight of knapsack
3	15	5	15-5=10
2	10	3	10-3=7
6	9	3	7-3=4
5	8	1	4-1=3
4	7x(3/4)=5.25	3	3-3=0

- **Total profit = 47.25**

Feasible Solution: 2

- **Choose the items according to their minimum weight**

Objects	Profits	Item weights	Remaining weight of knapsack
1	5	1	15-1=14
5	8	1	14-1=13
7	4	2	13-2=11
2	10	3	11-3=8
6	9	3	8-3=5
4	7	4	5-4=1
3	15x(1/5)=3	1	1=1=0

- **Total profit = 46**

Feasible Solution: 3

- **Choose the items according to their maximum ratio of profit by weight**

- Pi/wi:

5	3.3	3	1.75	8	3	2

Objects	Profits	Item weights	Remaining weight of knapsack
5	8	1	15-1=14
1	5	1	14-1=13
2	10	3	13-3=10
3	15	5	10-5=5
6	9	3	5-3=2
7	4	2	2-2=0

- **Total profit = 51**

Optimal Solution

- **A feasible solution is any set satisfying the objective of a problem but an optimal solution is a feasible solution for which the objective is maximized.**

- Therefore, in this case, feasible solution 3 gives the maximum profit. Thus, feasible solution 3 is the optimal solution for the fractional knapsack problem.

Pseudocode

Algorithm FractionalKnapsack(S, W):

Input: Set S of items, such that each item $i \in S$ has a positive benefit b_i and a positive weight w_i; positive maximum total weight W

Output: Amount x_i of each item $i \in S$ that maximizes the total benefit while not exceeding the maximum total weight W

for each item $i \in S$ **do**
  $x_i \leftarrow 0$
  $v_i \leftarrow b_i / w_i$ {***value index*** of item i}
$w \leftarrow 0$ {total weight}
while $w < W$ **do**
  remove from S an item i with highest value index {greedy choice}
  $a \leftarrow \min\{w_i, W - w\}$ {more than $W - w$ causes a weight overflow}
  $x_i \leftarrow a$
  $w \leftarrow w + a$

Time Complexity

- In the solution we use a heap-based priority queue to store the items of S, where the key of each item is its value index
- With priority queue, each greedy choice, which removes an item with the greatest value index, takes O(log n) time
- The fractional knapsack algorithm can be implemented in time O(n log n).